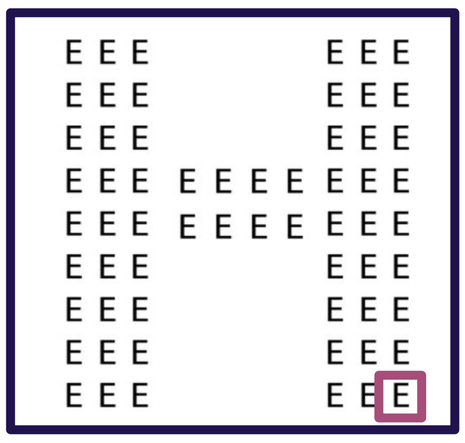
Predicting Words in Context: Children with and without Autism

Introduction

Weak Central Coherence (WCC) Theory

- Proposes autistic people focus on details (local) at the expense of overall context (global) [1,2].
  - Attributed to information integration difficulties.
- Proposed as domain-general cognitive processing style [1,2].



H → Global Form

E's → Local Detail

Prior Work:

- Strong support for WCC in the visual processing domain [3,4].
  - Role in language processing is less established [5].
- Autistic people rely more on local detail, rather than global form for both:
  - Homograph-Disambiguation [6].
  - Sentence Completion Tasks [1].

***Limitations*:**

- Prior work has not accounted for structural language ability as predictor of performance.
  - Language impairments often co-occur with autism [5].
- Limited set of stimuli tested.
- Recent changes in diagnostic criteria.

Research Questions

In a sentence completion task…

1) Do autistic children prioritise the immediate context (local detail) over the overall context (global form) compared to NT children?
   a) Extending to a new population and using new stimuli.
2) If performance differences are found, do they relate to autism diagnosis, language ability, or both?
   a) Extending prior work by considering structural language ability.

Methods & Materials

Sentence Completion Task: over Zoom, participants listened to sentence stems and were prompted to guess the final word.

Long and conflict sentences were designed to be the same length.

Sentence Types				
Condition	*N*	*Description*	*Example Sentence*	*Contextual Fit*
Short Sentences	15	One predictable completion.	The rabbit ate a _____.	Carrot
Long Sentences	15	One predictable completion.	The photographer had a big photoshoot [...] as the hours went by he got hungry. [...] He sat down and took out his _____.	Lunch
Conflict Sentences	15	Each clause suggests different predictable completions.	The Adams family decided to go on a road trip, [...] **The baby slept quietly in his _____.**	Carseat/ **Crib**

Sentences were presented in randomised order for participants.

Language Ability:

Clinical Evaluation of Language Fundamentals-4 (CELF-4)

- Broad test of children's language ability, produces a standard score [7]

Main Finding

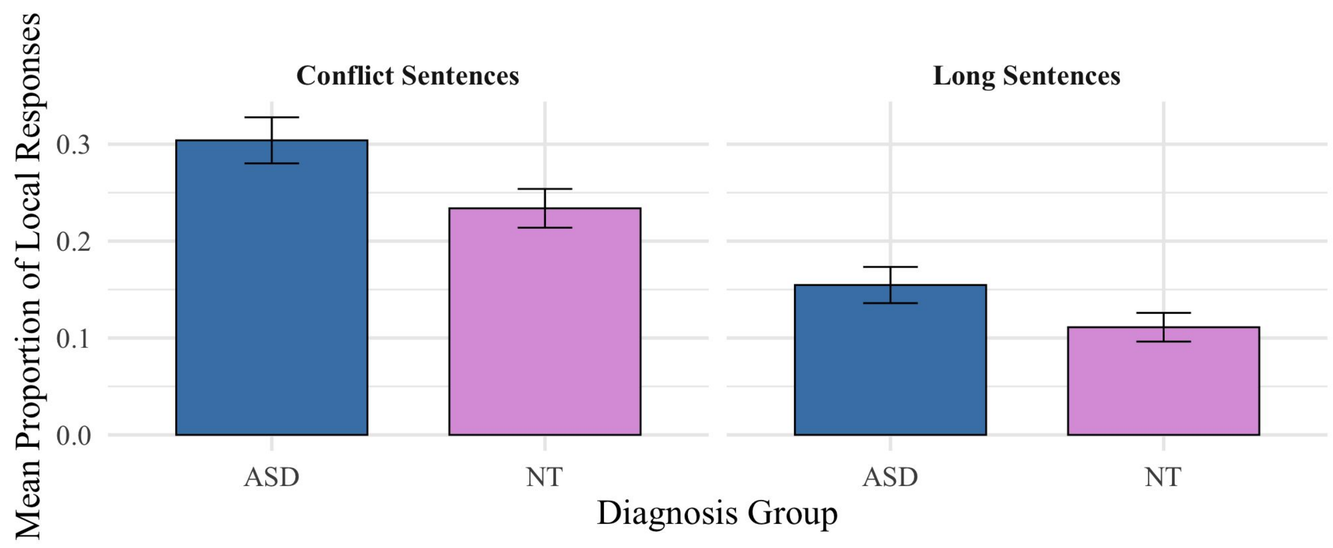
Language ability, not autism diagnosis, is the strongest predictor of differences in spoken language context use between autistic and neurotypical children in a sentence completion task.

Participants

	Autistic Children	Neurotypical Children
N after exclusions:	**24**	**30**
Excluded:	Incomplete: 4 Insufficient language proficiency: 1	Incomplete: 4
Age:	*R*= 7;5-12;11, *M*= 9;5	*R*= 6;2-10;11, *M*= 8;5
Language Score: *(CELF-4 average = 100)*	*R*= 58-111, *M*= 92.7	*R*= 90-132, *M*= 113.7

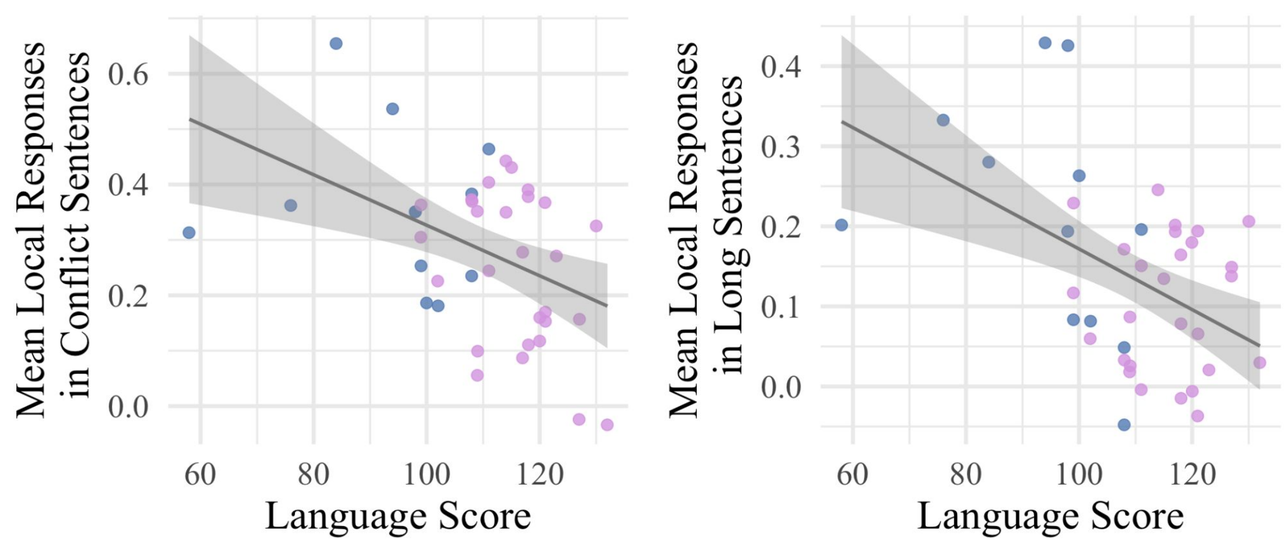
Results

1) Do autistic children prioritise the immediate context (local detail) over the overall context (global form) compared to NT children?



- **ASD children did not significantly produce more local responses** (β = -0.37, *SE* = 0.28, *p* = 0.18).
- **Conflict sentences had significantly more local responses overall** (β = 1.17, *SE* = 0.58, *p* = .044).
- **No interaction between sentence type and diagnosis** (β = 0.19, *SE* = 0.32, *p* = 0.56).

2) If performance differences are found, do they relate to autism diagnosis, language ability, or both?



- **Sentence type (long or conflict), no longer a significant predictor** (β = -0.01, *SE* = 1.57, *p* = 0.99).
- **Diagnosis status is not a significant predictor** (β = -0.25, *SE* = 0.33, *p* = 0.46).
- **Language ability is the <u>only</u> significant predictor of performance** (β = -0.03, *SE* = 0.01, *p* < .01).
- **No interaction between sentence type and diagnosis or sentence type and language ability.**

Discussion and Implications

1) **There is no significant difference in production of local responses between autistic and neurotypical children.**
   i) Going against WCC in language domain.
2) **Language ability is the <u>only</u> significant predictor of performance**
   i) Suggests reconsideration of conclusions drawn from prior WCC work in language domain.

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References

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